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Symptom clusters in ME/CFS reflect distinct neuroimmune and autonomic pathophysiological mechanisms: a translational model

Lotte Habermann-Horstmeier^{1*}  and Lukas M. Horstmeier²

Abstract

Background Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating multisystem disease characterized by heterogeneous symptom patterns. Previous work suggested that specific symptoms tend to co-occur, pointing toward underlying biological mechanisms. This study aimed to empirically validate literature-based, hypothesis-driven symptom clusters and assess whether they reflect distinct neuroimmune and autonomic pathophysiological pathways.

Methods Symptom data from 748 adults with ME/CFS (≥ 20 years) participating in the APAV-ME/CFS study were analyzed. Symptoms were assigned to predefined mechanistic groups informed by current pathophysiological hypotheses. Exploratory and Confirmatory Factor Analyses, followed by Structural Equation Modeling (SEM), evaluated the coherence, distinctiveness, and hierarchical structure of each cluster. Robustness was tested using a stratified, randomized training dataset.

Results A coherent *Brain* factor (brain fog, sensory hypersensitivity, visual disturbances, sleep disturbances, headaches) showed excellent fit (RMSEA = 0.021; CFI = 0.996). Gastrointestinal symptoms demonstrated stronger internal consistency than *Immune* symptoms, and model comparisons supported a two-factor *Gut–Immune* structure. Across all analyses, symptom groups emerged as internally consistent and statistically distinct. A higher-order SEM including a common latent factor yielded excellent fit for the *Autonomic* symptom complex.

Conclusions The findings support ME/CFS as a complex neuroimmune–autonomic multisystem disorder and suggest that symptom clusters align with functional biological systems. Mechanism-aligned symptom subgrouping may enable pathophysiology-guided diagnostics, patient stratification, and targeted therapeutic development. The proposed interpretations of underlying mechanisms derive from the integration of existing literature and were not directly measured in this study. The identified clusters therefore indicate mechanistic alignment rather than direct mechanistic validation.

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Keywords Myalgic Encephalomyelitis/Chronic Fatigue Syndrome, Symptom clusters, Neuroimmune mechanisms, Autonomic dysfunction, Translational modelling, Factor analysis, Structural equation modeling

Introduction

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS; ICD-10-GM-2025: G93.3) is a debilitating multisystem disease characterized by a wide range of symptoms [1] and substantial impairment in daily functioning. In most cases, ME/CFS follows an infectious trigger and is associated with long-lasting disability and markedly reduced quality of life [2]. Current evidence indicates disturbances across multiple regulatory systems—including the nervous, endocrine, and immune systems—as well as impaired cellular energy metabolism and chronic low-grade inflammation, suggesting a breakdown of physiological homeostasis in several interconnected biological networks [3–7].

A hallmark feature of ME/CFS is post-exertional malaise (PEM), a delayed and prolonged exacerbation of symptoms following minimal physical or cognitive exertion. PEM is accompanied by profound reductions in physical and mental energy, rapid deterioration of overall functioning, and a broad array of fluctuating symptoms, including neurocognitive impairment, immunological disturbances, sleep dysfunction, and autonomic dysregulation [5].

Despite the multisystemic nature of ME/CFS, little is known about how symptoms cluster interact, or evolve over the course of the disease. In a previous exploratory study (Study 1), we identified preliminary patterns suggesting that certain symptoms co-occur more frequently than expected, pointing toward underlying mechanistic relationships [8]. These findings motivated the present study (Study 2), which aims to empirically test and validate these hypothesized symptom clusters using Exploratory and Confirmatory Factor Analyses and Structural Equation Modeling.

We hypothesized that distinct symptom groups would emerge that align with known pathophysiological pathways—particularly neuroimmune, autonomic, gastrointestinal, and central nervous system mechanisms. Identifying such mechanism-aligned symptom clusters may support more precise diagnostics, inform stratification strategies, and guide the development of targeted therapeutic approaches. Building on these findings, the subsequent phases of this research program (Studies 3–5; see Fig. 1) will investigate potential modifiers of symptom architecture, with a particular focus on sex and illness duration. Studies 3 and 4 will examine possible sex-related differences in the *Brain/Brain2* and *Autonomic* clusters (Study 3) as well as in the *Gut/Immune/Immune2* clusters (Study 4), relating these patterns to established pathophysiological processes in ME/CFS and

contributing to the development of an integrated pathophysiological model. Study 5 will investigate potential changes in ME/CFS symptom clusters over the course of the disease, aiming to classify these changes pathophysiologically while differentiating between sexes.

Methods

This study is based on data from the APAV-ME/CFS study. The predominantly qualitative APAV-ME/CFS study focused primarily on the doctor-patient relationship in ME/CFS from the perspective of patients and their relatives. In addition, a range of basic data on gender/sex, age, etc., as well as information on whether the participant has a medical diagnosis of ME/CFS was also collected (see below).

Sampling

A total of 1,238 people (ME/CFS patients and their close relatives) took part in the APAV-ME/CFS study. They were recruited through voluntary participation and snowball sampling. To this end, contact was made in summer 2022 with six ME/CFS patient organizations in Germany as well as the Fatigue Centre at Charité in Berlin and the Chronic Fatigue Centre for Young People at the Technical University of Munich. The organizations forwarded the call for study participation to ME/CFS patients and their relatives. Through them, the information reached other affected persons. Long-term or post-COVID patients who did not meet the Canadian or International Consensus Criteria for ME/CFS were not included in the study.

The recruitment strategy via patient organizations and snowball sampling may introduce selection bias. However, this approach is widely used in ME/CFS research due to the absence of population-based registries and the limited availability of clinically diagnosed cohorts.

Study participants

The present study (Study 2) includes data from the 748 adult participants in the APAV-ME/CFS study who had already been medically diagnosed with ME/CFS (>20 years; 608 ♀, 137 ♂, 3 non-binary). The gender distribution in our sample (≈81% women) is consistent with international epidemiological data reporting a female predominance of 70–80%. Importantly, the aim of this study was not to estimate population prevalence but to examine the internal structure of symptom clusters. Data from participating children and adolescents (up to 20 years of age), relatives of ME/CFS patients, and the diseased individuals who had not yet been medically

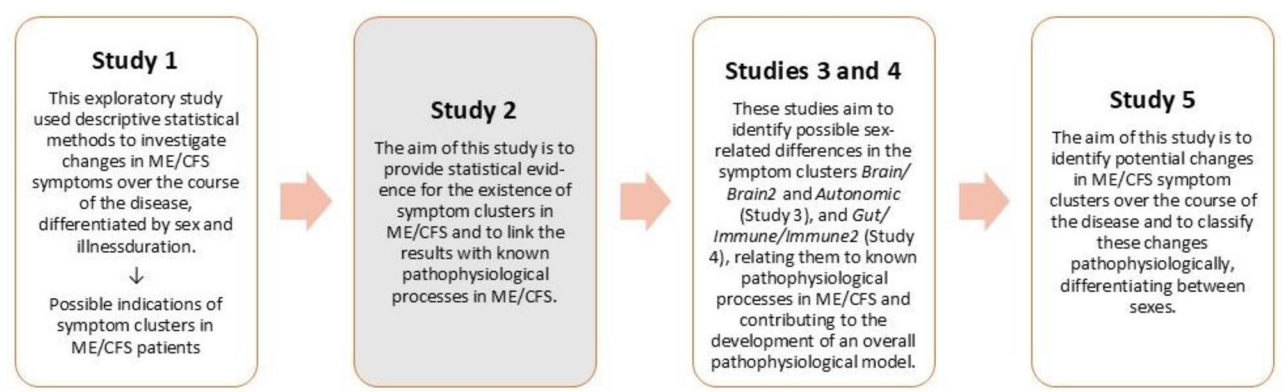


Fig. 1 Chronological placement of study 2 within the sequence of studies

Table 1 Description of the adult subjects (>20 years) with medically diagnosed ME/CFS, differentiated by gender and age group

Age groups	21–30 yrs.		31–40 yrs.		41–50 yrs.		51–60 yrs.		≥60 yrs.		Total	
	[n]	%	[n]	%	[n]	%	[n]	%	[n]	%	[n]	%
Women	51	8.4	115	18.9	183	30.1	197	32.4	62	10.2	608	100
Men	14	10.2	28	20.4	30	21.9	49	35.8	16	11.7	137	100
Non-binar	3	100.0	0	0.0	0	0.0	0	0.0	0	0.0	3	100
Total	68	9.1	143	19.1	213	28.5	247	33.0	78	10.4	748	100

Abbreviations: n = number of individuals, % = percentage, yrs. = years

diagnosed of ME/CFS were excluded. In addition, questionnaires lacking information on the relevant key variables – such as gender/sex, age, or conformation of medical ME/CFS diagnosis – were removed from the analysis (Table 1, Fig. 2).

Prior to participation, all respondents were informed about the study’s objectives, structure, and content. Participation was voluntary and anonymous, with no possibility of linking responses to individual identities. Informed consent was obtained through completion of the online questionnaire. To ensure appropriate inclusion, participants provided basic demographic and diagnostic information or symptom-related self-reports.

Study implementation and data processing

Almost all respondents completed the questionnaire we developed online (n = 1,118). The questionnaire consisted of a short quantitative and a more extensive qualitative section. In addition, the questionnaire was also available for download, enabling an additional 120 participants to return the completed questionnaire by email, fax, or post. The quantitative part of the questionnaire primarily collected basic information through closed questions, for example on the respondents’ age and gender. Additional questions include illness duration (self-assessed), the presence of a medical diagnosis, the date of diagnosis, the medical specialty and number of physicians consulted, the number of hospital and rehabilitation stays, the predominant ME/CFS symptoms, and whether symptoms improved or worsened following these hospitalizations.

Data analysis

All reported symptoms were coded as dichotomous variables (presence vs. absence). Dichotomizing symptoms into yes/no categories reduces variance and can attenuate factor loadings, internal consistency (Cronbach’s α), and SEM fit indices, particularly in heterogeneous domains such as the Immune and Autonomic clusters. This methodological constraint likely contributed to the moderate loadings observed in these domains. Nevertheless, global model fit indices remained excellent, indicating that the underlying latent structure was robust despite reduced measurement sensitivity. In the subsequent statistical analysis of the symptoms regularly occurring in the participants - except PEM and fatigue, as these are the core diagnostic criteria for an ME/CFS diagnosis and must therefore be present in all patients - tests for bivariate associations were first conducted, including a check of internal consistency (cross-tabulations, Cramér’s V and tetrachoric correlations). The Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) were then used to examine whether the previously defined, hypothesis-driven and pathophysiologically based subgroups could be identified as statistically detectable clusters.

All analyses were based on the sample of 748 patients, providing a sufficient number of cases for exploratory and confirmatory factor analyses. Although it is methodologically desirable to perform EFA and CFA on separate subsamples, we opted for a combined analysis because the sample size is too small to be split. To assess the robustness of the findings, the same analyses were repeated on

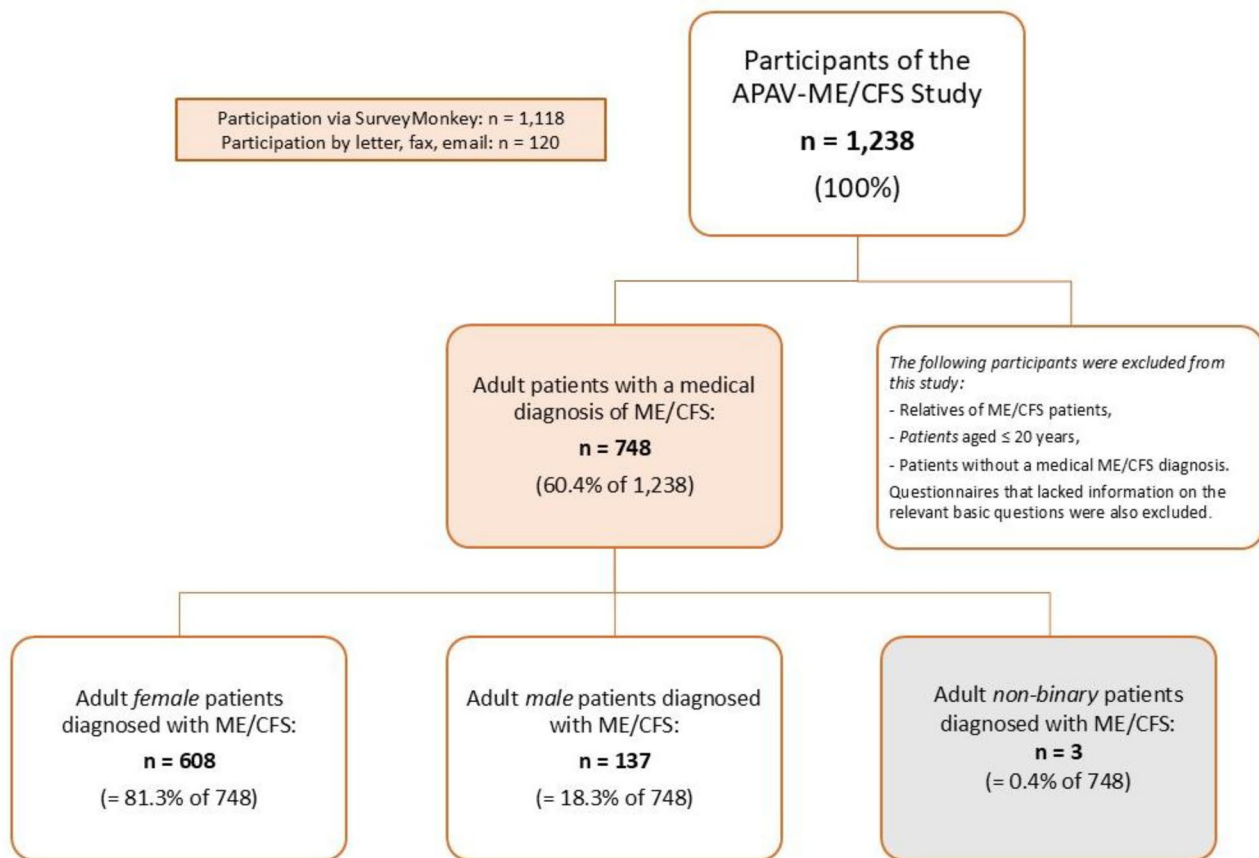


Fig. 2 Composition of the study sample

a stratified, randomized subsample (hereafter referred to as the training dataset) derived from the full sample¹. The training dataset was stratified by sex and illness duration to ensure that both variables were proportionally represented across all subsamples, and randomization within each stratum was applied to minimize potential biases and prevent systematic distortions. This procedure is particularly relevant in ME/CFS research, as sex and illness duration are known to influence symptom profiles and underlying pathophysiological mechanisms. The successful replication of all models in the stratified, randomized training dataset provides preliminary evidence for the structural stability of the identified clusters across demographic subgroups. Given that factor-analytic methods are comparatively robust to moderate sampling heterogeneity, this approach strengthens confidence in the stability and representativeness of the results and allows for cautious group-level comparisons.

All CFA and SEM models were originally estimated using maximum likelihood (ML). While ML is robust in large samples, it is not the optimal estimator for

dichotomous indicators. Categorical estimators such as WLSMV (Weighted Least Squares Means and Variance adjusted) are generally recommended for binary data. We therefore acknowledge this as a methodological limitation. Importantly, sensitivity checks using tetrachoric correlations and replicated analyses in the stratified training dataset yielded consistent factor structures, suggesting that the substantive conclusions are robust. Future phases of this research program will re-estimate all latent models using WLSMV to further strengthen methodological rigor.

Measurement invariance across sex, age groups, and illness duration could not be tested due to sample size constraints and the dichotomous nature of the indicators.

Cluster definition

In the following, hypothesis-led subgroups will be compared with the empirical-statistical symptom groups.

In other words, it should be shown,

- which theoretically defined clusters are statistically confirmed,
- where there may be deviations or overlaps and
- how new hypotheses or subtypes can be derived from this.

¹The results of these investigations can be viewed upon request from the corresponding author.

Table 2 shows the four hypothesized subgroups (*Brain*, *Autonomic*, *Gut* and *Immune*) including the most important pathophysiological mechanisms discussed in the literature.

The *Gut* and *Immune* clusters consisted of two indicators each. Two-indicator factors are statistically identified but do not allow meaningful evaluation of model fit. These constructs should therefore be interpreted as minimal representations of broader physiological domains. Their inclusion was justified by strong theoretical grounding in the literature, but future studies will incorporate additional indicators to improve reliability and model testability.

Some symptoms (e.g., gastrointestinal complaints and breathing difficulties) were defined a priori as potential bridging variables because prior literature suggests that they may involve more than one physiological subsystem (e.g., autonomic–enteric or autonomic–respiratory interactions). In the analytical framework, these symptoms were therefore permitted to load on more than one hypothesized latent domain. Their statistical behavior was examined both within clusters and across clusters to assess potential cross-domain associations and interface effects. This approach allowed us to evaluate whether these symptoms function as linking elements between functional subsystems. Their role and statistical characteristics are described in more detail in the subsequent sections, and will be examined further in a dedicated follow-up analysis within the broader study program to better characterize their potential bridging function across symptom domains.

Results

Neurocognitive-sensory symptom group (*Brain*)

For the statistical validation of the theoretically defined clusters, cross-tabulation analyses and calculations of Cramér’s V and tetrachoric correlations were first carried out in the area of the potential neurocognitive-sensory symptom group (*Brain*). An exploratory factor analysis was then carried out, followed by a Confirmatory Factor Analysis, which was used to test the fit of the *Brain* model to the data collected.

To improve readability, association metrics are summarized at the group level rather than interpreted for each symptom pair. Tetrachoric correlations consistently indicated medium-to-large latent associations across all clusters, and χ^2 tests were significant throughout

Bivariate associations and internal consistency

First, a neurocognitive-sensory symptom group (*Brain*) was defined, in which pathophysiological changes in the brain are assumed to be the central triggers for early-onset ME/CFS symptoms: (1) cognitive impairment (brain fog), (2) sensory hypersensitivity, (3) sleep disturbances, (4) visual disturbances and (5) headaches. Cross-tabulations and χ^2 -tests revealed significant associations, especially between cognitive impairment and sleep disturbances or sensory hypersensitivity as well as between visual disturbances and sensory hypersensitivity. The tetrachoric correlations (*r*) were consistently above the values of Cramér’s V and more adequately reflect the underlying continuous relationships. Almost all symptom pairs showed medium to high correlations ($r \geq 0.34$), indicating substantial latent coupling. The consistency of the

Table 2 Classification of the ME/CFS symptoms surveyed into functional clusters based on pathophysiological concepts

Subgroup	Symptoms	Possible pathophysiological mechanisms
Neurocognitive-sensory (<i>Brain</i>)	-Brain fog	-Neuroinflammation (microglia activation in the CNS) → cognitive dysfunctions
	-Sleep disorders	-Mitochondrial dysfunction and reduced cerebral blood flow → lack of energy in the brain
	-Sensory hypersensitivity	-Imbalance in the neurotransmitter balance (e.g. glutamate/GABA, serotonin) → sensory hypersensitivity & sleep disorders
	-Visual disturbances	-Dysfunction of the blood-brain barrier → Increased irritability of the nervous system
	-Headaches	-Autoantibodies against adrenergic and muscarinic receptors → Dysregulation of the autonomic nervous system
Autonomic Dysregulation (<i>Autonomic</i>)	-Cardiovascular problems	-Impaired baroreflex control → tachycardia, orthostatic intolerance
	-Temperature regulation disorders	-Decreased blood volume/hypovolemia → Circulatory problems
	-Respiratory problems	-Dysfunction of peripheral vascular regulation → Temperature and circulatory disorders
	-Gastrointestinal problems	Dysbiosis (change in the microbiome) → increased pro-inflammatory metabolites, reduced short-chain fatty acids
Gastrointestinal (<i>Gut</i>)	Gastrointestinal problems (Food) intolerances	"Leaky gut" (increased intestinal permeability) → endotoxemia, immune activation
		Enteric nerve dysfunction (as part of autonomic dysregulation) → Motility disorders
		Altered tryptophan and serotonin metabolism → gut-brain axis
Immunological (<i>Immune</i>)	Flu-like symptoms	Chronic low-grade immune activation (e.g. cytokine signature: IL-6, TNF-α, IFN-γ) → feeling ill, fatigue
	Susceptibility to infections	Exhausted NK cell function → reduced defense against viruses
		Persistent viral fragments (e.g. EBV, HHV-6, SARS-CoV-2) → permanent immune response
		Autoimmunity → against receptors and cellular structures

Literature sources see discussion

Table 3 Statistical correlations in the neurocognitive-sensory symptom group (*Brain*). Cross-tabulations and tetrachoric correlations

Pair of variables	N (total)	Tetrachoric ρ (SE)	χ^2 (df)	p-value
Brain fog ↔ Headache	748	0.4404 (0.067)	38.27	<0.0001
Brain fog ↔ Visual disturbances	748	0.5418 (0.061)	57.30	<0.0001
Brain fog ↔ Sleep disorders	748	0.5622 (0.065)	70.00	<0.0001
Brain fog ↔ Sensory hypersensitivity	748	0.5659 (0.063)	72.36	<0.0001
Visual disturbances ↔ Headache	748	0.3589 (0.055)	37.56	<0.0001
Visual disturbances ↔ Sleep disorders	748	0.3888 (0.064)	32.84	<0.0001
Visual disturbances ↔ Sensory hypersensitivity	748	0.5283 (0.055)	69.24	<0.0001
Headache ↔ Sleep disorders	748	0.3418 (0.067)	25.14	<0.0001
Headache ↔ Sensory hypersensitivity	748	0.4906 (0.057)	60.33	<0.0001
Sleep disorders ↔ Sensory hypersensitivity	748	0.4635 (0.066)	47.43	<0.0001

Abbreviations: N=number; tetrachoric ρ (SE)=tetrachoric correlation coefficient (standard error); χ^2 (df)=Chi² test (degrees of freedom); p-value=probability

Table 4a Confirmatory factor analysis (CFA) for the 'neurocognitive-sensory symptom group model' (*Brain*). Step 1: charges, residual variances and intercepts

Indicator	Factor loading λ	SE	z-value	p-value	95% CI for λ	Residual variance
Brain fog	1 (fixed)	-	-	-	-	0.0654
Sensory hypersensitivity	1.2522	0.1417	8.84	<0.001	[0.975, 1.530]	0.0911
Sleep disorders	0.9263	0.1091	8.45	<0.001	[0.712, 1.140]	0.0958
Visual disturbances	1.3846	0.1634	8.47	<0.001	[1.064, 1.705]	0.1741
Headache	1.1718	0.1512	7.75	<0.001	[0.875, 1.468]	0.1727

Abbreviations: λ =lambda; SE=standard error, p-value=probability; z-value measures how many standard deviations a data point is away from the mean; CI=confidence interval

results is confirmed by the consistency of the ranking of the association strengths in both metrics (Table 3).

Explorative factor analysis

The exploratory factor analysis (EFA) as a multivariate method for data reduction and structure identification serves to uncover latent factors behind the observed variables without a priori model specification (for details on the model specification and the eigenvalue criterion, see Box 1). The first eigenvalue was 1.20, all other eigenvalues were below 1.

Box 1: Model specification and eigenvalue criterion

Method

Principal factors on five binary symptoms:

-Brain fog

-Sensory hypersensitivity

-Headache

-Visual disturbances

-Sleep disorders

Retention according to Kaiser criterion (eigenvalue > 1): One factor

The likelihood ratio test against the independence model showed a significant deviation [χ^2 (10)=378.20, $p<0.0001$], which indicates substantial correlations between the items. The factor loadings on the first factor ranged from 0.43 to 0.55 and were thus classified as moderate. However, the uniqueness values of 0.70–0.82 showed that only 18–30% of the variance of the symptoms could be explained by the common factor. Since only one factor was extracted, the Varimax rotation did not lead to any change, as expected.

Table 4b Step 2: latent variance

	Value
Variance of "Brain"	0.0301
SE	0.0048
95% CI	[0.0230, 0.0411]

→ Factorized variance significant > 0

Confirmatory factor analysis

A Confirmatory Factor Analysis (CFA) within the Structural Equation Modeling (SEM) was used to estimate the strength of the relationships between the latent factor *Brain* and the observed variables (Box 2). The model was fitted using maximum likelihood estimation. Table 4 shows that all factor loadings were high and statistically significant ($p<0.001$). The residual variances of 0.06 to 0.17 prove that the factor explains a large part of the variance in the respective indicators. The latent variance of the *Brain* factor was estimated at 0.0301 (SE=0.0048). The 95% confidence interval was completely above zero, which indicates a significant variance of the factor.

Box 2: Model specification

One-factor model with latent variable Brain

Five binary indicators:

-Brain fog

-Sensory hypersensitivity

-Headache

-Visual disturbances

-Sleep disorders

Estimation with maximum likelihood (by default [ML])

Factor loading of Brain fog fixed to 1 to set the scale of the factor

Table 4c Step 3: global fit indices

Fit index	Value	Recommended limit value	Interpretation
χ^2 (df=5)	6.658	$p > 0.05$	Not significant → good fit
RMSEA	0.021	≤ 0.05	Excellent fit
90% CI RMSEA	[0.000, 0.058]	-	Tight interval
pclose	0.886	> 0.05	RMSEA ≤ 0.05 is plausible
CFI	0.996	> 0.95	Excellent fit
TLI	0.991	> 0.95	Excellent fit
SRMR	0.016	≤ 0.08	Excellent
CD (R ² equivalent)	0.645	-	Explains 64.5% of the total variance

Abbreviations: χ^2 (df)=Chi² test (degrees of freedom); p -value=probability; RMSEA=Root Mean Square Error of Approximation; CI=confidence interval; pclose=Probability RMSEA; CFI=Comperative Fit Index; TLI=Tucker-Lewis Index; SRMR=Standardized Root Mean Square Residual; CD=Coefficient of Determination; R²=statistical measure of how closely the data fit the fitted regression line

Table 5 Bivariate associations and reliability of autonomic symptoms

Variable pair	N (total)	Tetrachoric ρ (SE)	χ^2 (df=1)	p -value
Cardiovascular problems ↔ Breathing problems	748	0.605 (0.049)	101.25	< 0.001
Cardiovascular problems ↔ Temperature intolerance	748	0.620 (0.048)	110.75	< 0.001

Abbreviations: N=number; tetrachoric ρ (SE)=tetrachoric rho (standard error); χ^2 (df)=Chi² test (degrees of freedom); p -value=probability

We would like to point out that all analyses were conducted in Stata 12, which does not implement categorical SEM estimators such as WLSMV. Maximum likelihood estimation was therefore used for CFA/SEM models, supported by tetrachoric correlations and replication in a stratified training dataset. Although WLSMV would be preferable for dichotomous indicators, the convergence of results across analytic approaches suggests that the latent structure is robust. Future studies will employ software that supports categorical estimators to further refine the modeling framework.

In summary, the high and significant loadings (Table 4) confirm that the symptoms brain fog, sensory hypersensitivity, visual disturbances, sleep disturbances and headaches represent a coherent construct *Brain*. The low residual variances indicate that the latent factor explains a high proportion of the symptom variance. The global fit indices (non-significant χ^2 , RMSEA=0.021, CFI=0.996, SRMR=0.016) indicate a very good model fit.

Symptom group Autonomic dysregulation (*Autonomic*)

Table 6 Principal factors of the four autonomic indicators

Variable	Load weight λ to factor 1	Uniqueness	Eigenvalue factor 1	Explained variance (%)
Cardiovascular problems	0.585	0.658	1,153	28,8%
Breathing problems	0.553	0.716		
Temperature incompatibility	0.547	0.701		
Gastrointestinal complaints	0.477	0.773		

LR test: χ^2 (6)=370.2, $p < 0.001$ (against independence assumption)

Table 7 Predictors for cardiovascular problems (0/1) - logistic regression

Predictor	β (SE)	z	p	95% CI (β)	OR [95% CI]
Breathing problems	1.466 (0.226)	6.49	< 0.001	[1.024, 1.909]	4.33 [2.79, 7.10]
Temperature incompatibility	1.521 (0.223)	6.82	< 0.001	[1.084, 1.958]	4.58 [2.96, 8.03]
Gastrointestinal complaints	0.703 (0.226)	3.11	0.002	[0.259, 1.146]	2.02 [1.30, 3.14]
Constant	-0.606 (0.193)	-3.14	0.002	[-0.983, -0.228]	-
Pseudo R ²	0.233				

Bivariate associations and internal consistency

Vegetative symptoms showed substantial associations, with tetrachoric correlations between 0.48 and 0.62 (all $p < 0.001$). These values indicate a coherent latent structure despite expected heterogeneity. See Table 5 for details."

Explorative factor analysis

A one-factor solution on four *Autonomic* indicators (cardiovascular, respiratory problems, temperature intolerance, gastrointestinal complaints) resulted in a single eigenvalue > 1 (eigenvalue = 1.153), which explains 28.8% of the total variance. The unweighted factor loadings ranged from 0.477 to 0.585, uniqueness values were between 0.658 and 0.773. The factor analysis thus supports a unidimensional autonomic construct (*Autonomic*; Table 6).

Logistic regression

In a logistic regression model for the prediction of cardiovascular problems (0/1)² by breathing problems, temperature intolerance and gastrointestinal complaints, the values summarized in Table 7 were found. Respiratory problems and the intolerance of hot and cold (outside)

²The notation "(0/1)" indicates that the target variable is dichotomous—i.e., it has only two possible values—and that the logistic regression aims to model the probability of the value 1 (occurrence of problems).

temperatures multiply the odds for cardiovascular problems by a factor of 4 to 4.5, gastrointestinal complaints by a factor of 2. The pseudo- R^2 according to McFadden is 0.233, indicating a reasonably good model fit compared to a null model.

Confirmatory factor analysis

Structural Equation Modeling (SEM) with a common latent factor *Autonomic* for all four autonomic indicators provided excellent fit indices. The factor loadings were high and significant (Table 8). The residual variances ranged from $\theta=0.0879$ to $\theta=0.1600$, and the latent variance of the factors was $\phi=0.0617$ (SE = 0.0085). Proposed covariances between residuals of cardiovascular and gastrointestinal problems and between respiratory problems and temperature intolerance could not be confirmed either substantively or statistically.

Gastrointestinal symptom group (Gut) and immunological symptom group (Immune)

To test the dimensional structure of gastrointestinal and immunological symptoms, three structural equation models (SEM) were estimated: a one-factor model for gastrointestinal and immune-related symptoms, respectively, and a two-factor model with correlated latent variables. The most important parameters and fit indices are summarized in Table 9.

Both Cramér’s V and tetrachoric correlations were calculated to assess the association within the *Gut* (gastrointestinal complaints, (food) intolerances) and *Immune* (susceptibility to infections, flu-like symptoms) symptom groups. With a Cramér’s V of 0.405 and a tetrachoric correlation of $r=0.614$ (SE = 0.044), the *Gut* group showed a significantly stronger association between the symptoms than the *Immune* group ($V=0.294$; $r=0.505$, SE = 0.053). These results suggest a higher symptomatic coherence within the *Gut* symptom group (Table 10a). While Cramér’s V as a dimensionless measure describes the strength of the association between two categorical variables, the tetrachoric correlation allows an estimation of the relationship between the underlying latent constructs under the assumption of a bivariate normal distribution. The parallel consideration of both measures thus provides complementary evidence for the structural coherence of the respective symptom groups.

In addition, internal consistency was determined as a measure of the reliability of the groups using Cronbach’s alpha. A higher value of $\alpha=0.575$ was found for the *Gut* group than for *Immune* ($\alpha=0.450$). Although both values are below the threshold value of 0.70 often used as a guideline in psychometric studies, lower reliabilities are often to be expected with clinical symptom data due to great heterogeneity and are nevertheless meaningful. The combination of effect sizes and consistency scores thus

Table 8 SEM model with a latent-variable factor for autonomic symptoms and fit indices for a common factor *autonomic*

Indicator	Load weight λ	SE	Residual variance (SE)	Latent variance (SE)
Cardiovascular problems	1 (fixed)	-	0.0879 (0.0073)	0.0617 (0.0085)
Breathing problems	1.056	0.109	0.1536 (0.0105)	
Temperature incompatibility	1.048	0.106	0.1352 (0.0096)	
Gastrointestinal complaints	0.883	0.103	0.1600 (0.0099)	
χ^2 (2)			TLI	SRMR
5.38			0.972	0.015
			CFI	AIC
			0.991	3237.65
				BIC
				3293.06

Abbreviations: λ = Lambda; SE = Standard error, χ^2 (df) = Chi² test (degrees of freedom); p-value = Probability; RMSEA = Root Mean Square Error of Approximation; CI = Confidence Interval; pclose = Probability RMSEA; CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; SRMR = Standardized Root Mean Square Residual; AIC = Akaike information criterion, BIC = Bayes information criterion

Table 9 Comparison of three structural equation model variants. Models A and B are separately estimated one-factor models, model C is a jointly estimated two-factor model with correlated latent variables

	Separate 1-factor model A	Separate 1-factor model B	2-factor model C
Symptom group	<i>Gut</i> (gastrointestinal group)	<i>Immune</i> (Immune Group)	<i>Gut & Immune</i>
Factor	Latent factor <i>Gut</i>	Latent factor <i>Immune</i>	Common factor for all four symptoms
Indicators	Gastrointestinal problems, (Food) intolerances	Flu-like symptoms Susceptibility to infections	All four symptoms
Factor loading	(Food) intolerances: 0.43	Infektanfälligkeit: 0.74	(Food) intolerances: 1.00 (fixed) Gastrointestinal problems 0.91 Flu-like symptoms 0.54 Susceptibility to infections 0.74

supports the assumption that the *Gut* symptom group represents a more homogeneous sub-construct within the ME/CFS symptom spectrum than the *Immune* symptom group (Table 10b).

The separately estimated models converged without any problems and yielded plausible factor loadings. In particular, the model for immune-related symptoms (susceptibility to infection, flu-like symptoms) showed a high factor loading of $\lambda = 0.74$ with simultaneously low residual variance, which indicates a strong link between the indicators and the underlying factor.

In contrast, the joint one-factor model showed a significantly poorer model fit ($\chi^2(2) = 23.78, p < 0.001$), indicating that the joint variance of all four symptoms cannot be adequately explained by a single latent factor. The results therefore clearly support a differentiated modeling, i.e. the symptoms should be recorded in two separate groups - a gastrointestinal group (*Gut*) consisting of gastrointestinal complaints and (food) intolerances and an immunological group (*Immune*) consisting of susceptibility to infections and flu-like symptoms.

At the same time, the cross-tabulation of GutScore and ImmuneScore reveals a significant association between gastrointestinal and immune-related symptoms ($\chi^2(4) = 89.24, p < 0.001$). This indicates that many participants exhibit symptoms in both domains simultaneously. For example, among those with high GutScore (=both *Gut* symptoms are present), more than half (54.4%) also show high ImmuneScore (=both *Immune* symptoms are

Table 10a Statistical correlations in the two symptom groups *Gut* and *Immune*. Cross-tabulations, Cramér's V and tetrachoric correlations

Variable pair	N (total)	χ^2 (df)	p-value	Cramér's V	Interpretation V according to Cohen	Tetrachoric r	Interpretation r according to Cohen	Cronbach's Alpha	Item covariance
<i>Gut</i> : Gastrointestinal problems ↔ (Food) intolerances	748	122.54 ¹	<0.001	0.405	Moderate to strong (clear) correlation	0.614 (SE = 0.044)	strong effect	0.575	0.090
<i>Immune</i> : Flu-like symptoms ↔ Susceptibility to infections	748	64.720 ¹	<0.001	0.294	almost moderate correlation	0.505 (SE = 0.053)	strong effect	0.450	0.063

Table 10b SEM analysis

Model	Indicator	Factor loading λ (SE)	Residual variance (SE)	Latent variance (SE)	χ^2 (df), p	RMSEA (90% CI), pclose	CFI	TLI	SRMR
<i>Gut</i> (1-factor)	Gastrointestinal complaints	1 (fixed)	≈0*	0.2391 (0.0233)	– (df = 0)		1.000	1.000	
	(Food) intolerances	0.4338 (0.0179)	0.19995 (0.01083)						
<i>Immune</i> (1-factor)	Flu-like symptoms	1 (fixed)	≈0*	0.2491 (0.0323)	– (df = 0)		1.000	1.000	
	Susceptibility to infection	0.3448 (0.0183)	0.22758 (0.01204)						
<i>Gut & Immune</i> (2-factors)	Susceptibility to infection	1 (fixed)	0.19934 (0.01269)	0.04342	23.78(2), ≤0.000	0.121 [0.080, 0.166], 0.003	0.923	0.769	0.038
	Flu-like symptoms	0.7330 (0.1063)	0.15459 (0.00912)						
	Gastrointestinal complaints	1.2327 (0.1722)	0.13250 (0.11057)						
	(Food) intolerances	1.3526 (0.1878)	0.14803 (0.01298)						
Covariance <i>Gut</i> – <i>Immune</i>	0.03792 (0.00675)								

*Fixed charge leads to arithmetically vanishing residual variance

Abbreviations: N = number; χ^2 (df) = Chi² test (degrees of freedom); p-value = probability; Cramér's V = variable V, which quantifies the statistical relationship between two nominally scaled variables; Tetrachoric r = Tetrachoric correlation coefficient, RMSEA = Root Mean Square Error of Approximation; CI = Confidence Interval; pclose = Probability RMSEA; CFI = Comparative Fit Index; TLI = Tucker-Lewis Index; SRMR = Standardized Root Mean Square Residual

present), suggesting a frequent co-occurrence of these symptom clusters.

Special features of the symptoms ‘gastrointestinal complaints’ and ‘breathing difficulties’

Two of the symptoms ticked by the participants (gastrointestinal complaints and breathing difficulties) could each be assigned to two symptom groups, so they play a special role.

Gastrointestinal complaints

Gastrointestinal complaints belong to the *Autonomic* subgroup, but together with (food) intolerances they define the *Gut* subgroup. The results of the factor analysis in the *Autonomic* subgroup (Table 8) show that gastrointestinal complaints only load moderately on the general autonomic factor ($\lambda = 0.477$; Uniqueness = 0.773). At the same time, the two-factorial Structural Equation Modeling (SEM; Gut vs. Immune, Table 10b) resulted in a factor loading of $\lambda = 1.00$ for gastrointestinal complaints in the pure *Gut* model, while (food) intolerances were associated with $\lambda = 0.434$. The bivariate associations also support these findings. Thus, gastrointestinal complaints correlated significantly with other autonomic indicators (e.g. Cramér's V: 0.37–0.39; tetrachoric $r \approx 0.62$). On the one hand, they reflect the generalized autonomic dysregulation; on the other hand, the Gut construct forms an independent, relatively homogeneous dimension of gastrointestinal phenomena (Cronbach's $\alpha = 0.575$; inter-item covariance = 0.090), which will be discussed further in the discussion and in the planned fourth part of our five-stage study series.

Breathing problems

The statistical analyses (cross-tabulations, association and reliability measures), which are not explained in detail here, show a moderate association between muscle and breathing problems (Cramér's V = 0.292; $\rho = 0.530$), which indicates a clear, but not very strong, co-occurrence of these symptoms. At the same time, there is a low internal consistency (Cronbach's $\alpha = 0.433$), which is below established threshold values for scale homogeneity ($\alpha \geq 0.70$). This means that muscle and breathing problems form a partially overlapping but not uniform dimension. At the same time, the symptom “breathing difficulties” is part of the autonomic subgroup, which shows in Table 8 that breathing difficulties load moderately on the general autonomic factor ($\lambda = 0.553$; uniqueness = 0.716). Possible pathophysiological correlations underlying this will be discussed in a later publication.

Brief summary of the statistical results

Highly significant correlations were found within the previously pathophysiological defined symptom

subgroups. The statistical calculations confirmed symptom clusters in the following areas:

- neurocognitive
- sensory
- autonomic
- gastrointestinal
- immunological

The fact that the neurocognitive-sensory symptoms in particular represent a coherent *Brain* construct was shown above all by the high and significant loadings in the factor analysis. At the same time, low residual variances indicated that the latent factor explains a high proportion of the symptom variance. In addition, the global fit indices demonstrated a very good model fit. Structural Equation Modeling (SEM) with a common latent factor also provided very good fit indices for the *Autonomic* symptom complex. A comparison of the two one-factor models for the gastrointestinal symptoms (*Gut*) and the symptoms assigned to the immune group (*Immune*) with a joint two-factor model with correlated latent variables showed a significantly poorer model fit for the two-factor model. This supports the assumption of two clearly distinguishable but related latent constructs.

Comparison of hypothesis-driven symptom groups and empirically-statistically found symptom clusters

The aim of this study was to empirically test and validate hypothesized ME/CFS symptom groups using statistical methods. Table 11 summarizes the extent to which the four theoretically defined ME/CFS subgroups were confirmed by the statistical analyses performed and also indicates the identified overlaps (“bridging phenomena”). Bridging phenomena are defined here as those cases in which symptoms can be assigned to more than one functional subsystem, as they mediate both statistically and pathophysiologically between two subgroups. They therefore not represent pure markers of a single subgroup, but instead connect two dimensions, demonstrating that the subgroups are not strictly separated from one another but are functionally networked.

While cross-loading or bifactor models would be methodically ideal to formally capture such overlap, these models were not estimable with dichotomous indicators in the available software environment. In line with this interpretation, the poor fit of the joint one-factor Gut–Immune model ($\chi^2 = 23.78$, $p < 0.001$) further supports the assumption that these domains represent distinct but interconnected constructs. Future work will apply more flexible modeling approaches (e.g., ESEM, Bayesian SEM, or network models) to better account for symptom overlap.

Table 11 Extent to which the four theoretically defined ME/CFS subgroups were confirmed by the statistical analyses performed

Subgroup (theory)	Core symptoms	Empirical evidence	Key findings
<i>Brain</i> (neurocognitive-sensory)	- Brain fog- Sleep disorders- Sensory hypersensitivity - Visual disturbances- Headaches	- Bivariate associations: $r \geq 0.34$ – 0.56 (strong effects) - EFA: One factor (Eigenvalue = 1.20)- CFA: $\chi^2(5) = 6.66$, $p = 0.246$; RMSEA = 0.021; CFI = 0.996; SRMR = 0.016	Very coherent 1-factor construct, high loadings ($\lambda = 0.93$ – 1.38), low residual variances
<i>Autonomic</i> (autonomic)	- Cardiovascular problems - Breathing problems - Temperature intolerance - Gastrointestinal complaints	- Cramér's $V \approx 0.37$ – 0.39 ; $r \approx 0.60$ – 0.62 - EFA: 1 Faktor, 28.8% Varianz - SEM (Autonomic): $\chi^2(2) = 5.38$, $p = 0.068$; RMSEA = 0.047; CFI = 0.991; SRMR = 0.015	Unidimensional autonomic scale with high loadings ($\lambda = 0.88$ – 1.06) and good model fit
<i>Gut</i> (gastrointestinal)	- Gastrointestinal complaints - (Food) intolerances	- Cramér's $V = 0.405$, $r = 0.614$ - SEM (1-Faktor): $\lambda_1 = 1.00$; $\lambda_2 = 0.434$; $\phi = 0.208$; perfectly identified	Homogeneous gastrointestinal construct, high symptomatic coherence ($\alpha = 0.575$)
<i>Immune</i> (immunological)	- Flu-like symptoms - Susceptibility to infections	- Cramér's $V = 0.294$, $r = 0.505$ - SEM (1-Faktor): $\lambda_1 = 1.00$; $\lambda_2 = 0.345$; $\phi = 0.181$; perfectly identified	Stand-alone immune construct with moderate internal consistency ($\alpha = 0.450$)
„Bridge phenomena“ Gastrointestinal complaints are carriers of the pure good factor ($\lambda = 1.00$), but at the same time load moderately on the autonomic factor (<i>Autonomic</i> ; $\lambda = 0.477$). Although breathing problems correlate with muscle and autonomic symptoms, they do not form a uniform scale ($\alpha = 0.433$).			

Discussion

The present study represents the second phase of a multi-stage research program aimed at elucidating the symptom architecture of ME/CFS. Building on preliminary findings from Study 1, which suggested sex-specific patterns and co-occurrence of symptoms [8], this study sought to empirically validate predefined, hypothesis-driven symptom clusters using factor analytic methods and structural equation modeling (SEM). Across a large cohort of clinically diagnosed ME/CFS patients, we identified four robust and internally coherent symptom domains—*Brain*, *Immune*, *Autonomic*, and *Gut*—that align closely with current pathophysiological hypotheses. Replication in a stratified, randomized training dataset confirmed the stability of these clusters, underscoring their structural validity. It is important to emphasize that the biological mechanisms discussed here were not directly assessed in this study. Mechanistic interpretations are inferred from converging evidence in the literature and should be understood as alignment rather than empirical validation.

Symptom clusters reflect distinct neuroimmune and autonomic mechanisms

Brain cluster

The *Brain* cluster, comprising neurocognitive, sensory, and sleep-related symptoms, is consistent with extensive evidence of central nervous system abnormalities in ME/CFS. Neuroimaging studies have demonstrated structural and functional alterations in the brainstem, thalamus, and prefrontal cortex [9–14], including reduced cerebral blood flow [15], neuroglial dysfunction [13], and widespread neuroinflammation [14]. Altered sleep architecture has also been documented [16]. These findings

support the interpretation of the *Brain* cluster as a manifestation of network-level dysregulation within thalamo-cortical and brainstem circuits.

Immune cluster

The *Immune* cluster aligns with well-established immunological abnormalities in ME/CFS, including impaired NK-cell function [17], genetically inferred immune dysregulation [18], chronic activation of pro-inflammatory cytokines [18, 19], and reactivation of latent herpesviruses such as HHV-6 [20] and EBV [21]. These processes contribute to sickness-behavior-like symptoms [22], neuroinflammation [4, 23], and increased susceptibility to infections. The coherence of this cluster reinforces the role of immune dysregulation as a central biological driver of symptom expression.

Autonomic cluster

The *Autonomic* cluster reflects autonomic dysfunction across cardiovascular, respiratory, and thermoregulatory domains. Reduced heart rate variability, orthostatic intolerance, tachycardia, and chronotropic incompetence have been repeatedly documented [24, 25]. Dysregulation of vagal pathways may contribute to nausea and delayed gastric emptying [26], while altered chemoreceptor and baroreceptor function may influence respiratory rate, CO₂ regulation, and cerebral perfusion [15]. Thermoregulatory instability and vascular tone dysregulation have also been reported [27]. These findings support the conceptualization of ME/CFS as a disorder of integrated neuroimmune–autonomic regulation [28].

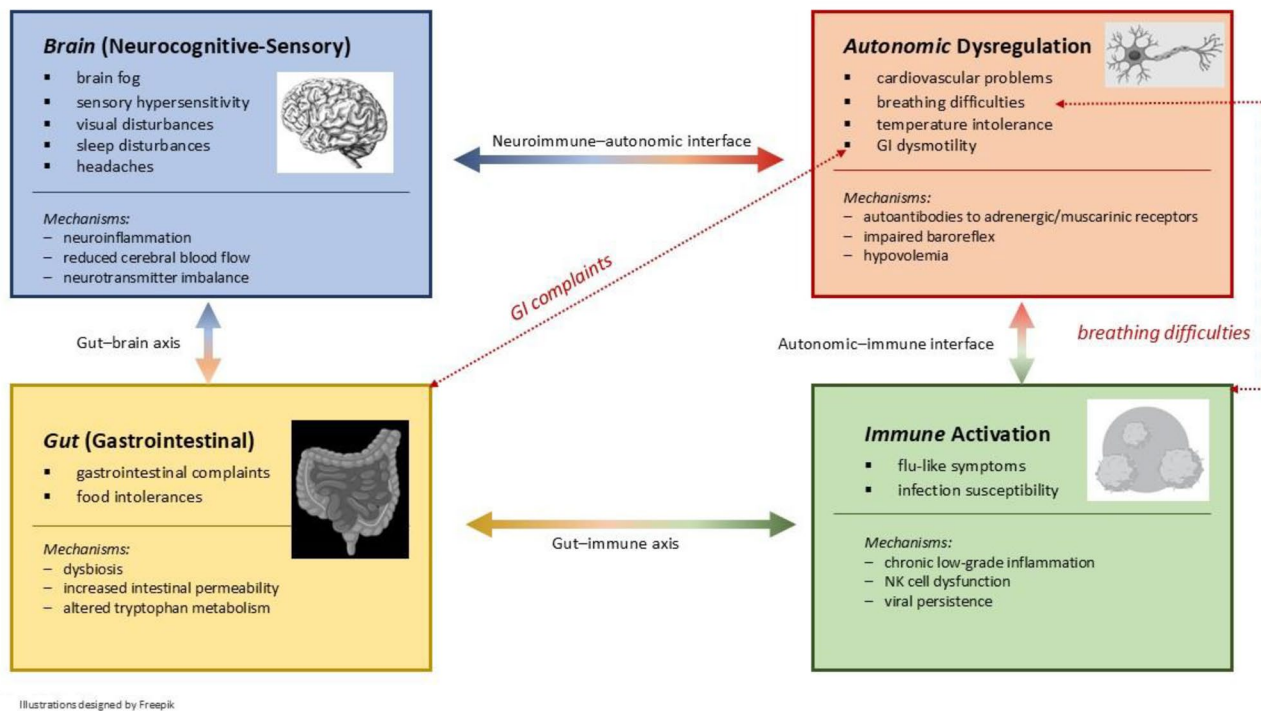


Fig. 3 Translational model of ME/CFS symptom architecture. The model depicts four pathophysiologically aligned symptom clusters (Brain, Autonomic, Gut, Immune), their underlying pathophysiological mechanisms, and the physiological axes linking them (neuroimmune–autonomic, gut–brain, autonomic–immune, and gut–immune). Symptoms positioned at cluster interfaces illustrate overlapping functional domains and highlight sites of inter-system interaction

Gut cluster

The *Gut* cluster captures gastrointestinal symptoms that extend beyond autonomic dysregulation and reflect distinct enteric pathomechanisms. Dysbiosis, reduced microbial diversity, and altered microbiota composition have been documented in ME/CFS [29, 30], alongside increased intestinal permeability and bacterial translocation [31, 32]. These abnormalities may activate immune pathways through the gut–brain–immune axis [33, 34]. The separation of *Gut* and *Immune* clusters, despite their functional interconnections, highlights the presence of partially independent but interacting biological subsystems.

Analyses conducted on the stratified, randomized training dataset yielded directionally consistent factor loadings and comparable fit indices to the full dataset. No structural deviations were observed, supporting the robustness of the identified clusters. Detailed results are available in the additional online media.

Overall model reflecting distinct neuroimmune and autonomic mechanisms

Together, these findings demonstrate that ME/CFS symptoms do not occur randomly but cluster in patterns that correspond to functional body systems and known biological pathways. The presence of “bridging symptoms”

across clusters underscores the interconnected nature of neuroimmune, autonomic, and enteric mechanisms. Our model integrates clinical symptom domains with convergent pathophysiological mechanisms to depict four mechanistically defined clusters (*Brain*, *Autonomic*, *Gut*, *Immune*). Each cluster represents a symptom constellation anchored in specific biological processes, including neuroinflammation and cerebral hypoperfusion (*Brain*), receptor-mediated autonomic dysfunction and impaired baroreflex control (*Autonomic*), microbiome alterations and intestinal barrier dysfunction (*Gut*), and chronic low-grade immune activation with NK cell dysfunction (*Immune*).

Interconnecting physiological axes (neuroimmune, gut–brain, autonomic–immune, and gut–immune pathways) illustrate bidirectional signaling and cross-system modulation. Symptoms located at the interfaces between clusters mark zones where different phenotypes overlap. They indicate shared mechanisms and reciprocal biological influences between the involved regulatory systems. In this way, the model makes visible how symptom structure functions as a systemic architecture that links molecular dysfunction to the clinical phenotype (Fig. 3).

Comparison with previous cluster and factor analytic studies

Our results are broadly consistent with earlier cluster and factor analyses, which also identified neurocognitive, autonomic, and immune-related symptom domains [6, 35–39]. However, the present study extends previous work in several important ways. First, we tested predefined mechanistic clusters derived from pathophysiological hypotheses rather than relying solely on data-driven approaches. Second, our sample is among the largest analyzed to date in ME/CFS cluster research. Third, replication in a stratified dataset strengthens the robustness and generalizability of the findings. The emergence of a distinct Gut cluster and the identification of bridging phenomena provide additional granularity that has been underrepresented in earlier studies.

Implications for diagnostic refinement and translational research

The identified clusters have direct implications for diagnostic practice and translational research. Current diagnostic instruments, including CCC-based questionnaires [40] and widely used clinical tools in Germany [41], do not fully capture the biological architecture of ME/CFS. Several relevant symptoms—such as visual disturbances, muscle weakness, and gastrointestinal or respiratory bridging symptoms—are insufficiently represented or absent. Our findings suggest that diagnostic tools should incorporate mechanism-aligned subscales reflecting neurocognitive, immune, autonomic, and enteric domains. Such differentiation may improve diagnostic precision, support patient stratification, and guide individualized therapeutic strategies.

From a translational perspective, the clusters provide testable mechanistic hypotheses for biomarker development. Integrating symptom clusters with immune profiling [3, 17, 18], microbiome analyses [30–33], metabolic markers, and neuroimaging [9–15] could refine phenotype definitions and identify biological signatures associated with specific symptom domains. This approach may facilitate the development of targeted interventions and improve the design of clinical trials by enabling more homogeneous subgroup selection.

However, the diagnostic and stratification implications outlined here should be understood as conceptual and hypothesis-generating. Clinical implementation will require validation with biological markers, longitudinal data, and independent cohorts.

Recent biomarker studies have demonstrated immune activation and metabolic abnormalities not explicable by inactivity [3], while deep phenotyping studies have highlighted multisystem involvement [1]. Qualitative research further indicates that patients consistently describe biological triggers—most commonly viral infections—and

report frequent misdiagnosis in psychosomatic contexts [42, 43]. The present findings thus contribute to the growing body of evidence supporting a biological disease model.

Methodological considerations and future directions

Despite the excellent fit indices of our SEM models, several methodological limitations warrant consideration. Dichotomous symptom coding reduces variance and may obscure gradations in symptom severity. Measurement invariance across sex, age, and illness duration has not yet been assessed. Moderate Cronbach's α values are therefore not unexpected in multisystem conditions such as ME/CFS, where symptom domains are inherently heterogeneous. In this context, moderate internal consistency does not weaken the construct's validity, especially when it is supported by converging evidence from EFA, CFA, and SEM.

Differences in explained variance between EFA and CFA are expected because CFA applies a more constrained structure, which typically increases the proportion of variance attributed to the latent factor. These differences do not indicate inconsistency but reflect the distinct purposes of exploratory versus confirmatory modeling. Importantly, the biological mechanisms discussed in this manuscript were not directly measured. They represent literature-based interpretations aligned with the observed symptom structure rather than empirical validation.

Future studies should employ polytomous symptom scales, incorporate invariance testing, and validate the identified clusters in larger, independent, and ideally longitudinal cohorts. Epidemiological studies indicate age-related incidence peaks [44] and substantial disease burden [2], underscoring the need for refined phenotyping across demographic subgroups. The next phases of this research program will therefore examine how symptom clusters vary by sex, age, and disease duration and explore their associations with biological markers. Such work will be essential for advancing precision-medicine approaches and developing targeted preventive and therapeutic strategies.

Limitations

Participants were recruited via ME/CFS self-help groups and the two German treatment centers in Berlin and Munich. As recruitment was based on self-selection and snowballing, selection and homogeneity effects may have occurred, but with a sample size of $n=748$ these can be considered moderate. A comparison with the actual gender- and age-related prevalence/incidence in Germany was not possible due to a lack of reliable data. Women are likely slightly overrepresented (611 women vs. 138 men), but prevalence and incidence peaks correspond to

international findings (incidence peak: 30–39 years; prevalence peak: 40–59 years) [44, 45].

Participants' medical diagnoses and symptoms were self-reported and did not rely on clinical verification. This may introduce selection and diagnostic bias. However, the aim of the study was not to estimate prevalence or evaluate diagnostic accuracy but to examine the internal structure of symptom clusters. Factor-analytic methods are comparatively robust to such sampling heterogeneity.

Self-report Data may also be affected by recall bias. In addition, the pathophysiological classifications discussed in this study are derived from existing literature, rather than from direct biological measurements. Given the limited sample size, exploratory and confirmatory factor analyses (EFA and CFA) could not be conducted on independent subsamples; instead, the full dataset was used. To assess the robustness of the findings, additional analyses were performed using a randomized, stratified training subsample stratified by sex and disease duration. While this procedure improves representativeness and reduces the risk of systematic bias, it does not replace validation in fully independent samples.

The dichotomization of symptoms (yes/no) further constitutes a methodological constraint, as binary indicators reduce variability and statistical power in structural equation modeling (SEM) and may attenuate factor loadings. This likely contributed to the moderate loadings observed in domains such as the *Brain* cluster. Dichotomous items are nevertheless common in clinical epidemiology because they reduce respondent burden and minimize recall bias. However, they capture severity gradients only indirectly. Despite this limitation, all CFA and SEM models demonstrated excellent global fit indices, indicating that the underlying latent structure is robust. Future phases of this research program will therefore employ polytomous symptom scales to capture severity more precisely and further strengthen measurement sensitivity. Finally, potential effects of sex, age, or disease duration on the latent factors were not examined in the present analyses. Likewise, formal tests of measurement invariance across demographic subgroups were not feasible. These aspects will be addressed in later phases of the research program.

The use of maximum likelihood (ML) estimation with binary indicators represents a methodological limitation. Although ML performs adequately in large samples, categorical estimators such as WLSMV are generally preferable for dichotomous data and will therefore be implemented in future analyses. This limitation does not substantially affect the overall interpretation of the latent structure but should be addressed in later phases of the research program.

In addition, we were unable to elegantly map local dependencies of certain symptoms (e.g. heart ↔ gut)

using the Stata 12 software available to us. As a result, parameters could be distorted or fit indices misleading. Multiple model comparisons risked alpha inflation. Thus, Bonferroni correction was applied. Data on psychiatric comorbidities, medication use, and socioeconomic status were not available in the APAV-ME/CFS dataset and could therefore not be included as covariates. Consequently, potential confounding factors could not be controlled, which may have influenced symptom reporting and biased the identified symptom clusters. Moreover, recorded psychiatric diagnoses may themselves represent misdiagnoses [42], further complicating the interpretation of these effects. Although these limitations must be acknowledged, the primary focus of this study was on the internal structure of symptom clusters rather than on causal inference, and factor-analytic methods are comparatively robust to such confounding influences. Future studies will incorporate these variables to allow more comprehensive modeling and to further refine the interpretation of symptom domains.

Conclusions

Exploratory and Confirmatory Factor Analyses, together with Structural Equation Modeling (SEM), consistently revealed distinct, internally coherent symptom clusters in ME/CFS that demonstrate excellent model fit and align with current neuroimmune and autonomic pathophysiological hypotheses. These findings strengthen the conceptualization of ME/CFS as a complex multisystem disorder and provide translationally relevant evidence that the symptom patterns observed here reflect the underlying biological mechanisms. The statistically validated clusters correspond closely to previous observations and provide a basis for mechanism-informed diagnostics and individualized therapeutic strategies. However, although the identified clusters align with the aforementioned neuroimmune and autonomic mechanisms, they do not directly validate them, as the biological signaling pathways were derived from previous research and were not measured in the present dataset. Furthermore, the *Gut* and *Immune* factors were defined by only two indicators, resulting in perfectly identified models that do not permit evaluation of global fit. This limits the reliability of these constructs and should be addressed in future work by expanding the item sets.

Our results also highlight the need for diagnostic instruments that more accurately reflect the biological architecture of ME/CFS. Future questionnaires should incorporate differentiated subscales—such as gastrointestinal, immune, neurocognitive, and autonomic domains—to better capture disease heterogeneity and guide targeted interventions. Additional symptom domains, including muscle and ocular symptoms, warrant inclusion as separate subscales.

Further research should replicate these clusters in larger, independent, and ideally longitudinal cohorts, while integrating biological markers such as immune signatures, microbiome profiles, metabolic parameters, and neuroimaging findings to refine mechanistic understanding. Analytic approaches should also be expanded to include machine-learning approaches (e.g., clustering algorithms, random forests, or network-based community detection) to complement factor-analytic techniques and explore potential nonlinear or higher-order symptom structures.

Despite the excellent fit indices of our SEM models, methodological limitations must be acknowledged. Dichotomous symptom coding reduces variance and may obscure subtle effects, and measurement invariance across sex, age, and illness duration has not yet been assessed. Future studies should therefore employ polytomous symptom scales and incorporate invariance testing to enhance robustness and generalizability. In addition, examining symptom clusters across demographic and clinical subgroups may support the development of tailored preventive and therapeutic approaches.

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Author's contributions

LHH contributed to the development and conception of the research project, the preparation, collection, procurement, and provision of the data, the analysis and interpretation of the data, the review of the sources, the conclusions drawn from these findings, and the writing of the manuscript. LMH contributed to the preparation, provision, and interpretation of the data.

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Data availability

All data supporting the results of this study are included in the article or can be requested from the corresponding author.

Declarations

Ethical approval

This study was conducted in 2022 in accordance with the Declaration of Helsinki and approved by the Ethics Committee of Furtwangen University, Germany (application number: 22–057).

Consent for publication

Not applicable.

Consent to participate & written consent for publication

All participants were fully informed about the objectives and procedure before the study began and gave their written consent to participate. See *Information and declaration of consent* in the "Supplementary files" section.

Code availability

Not applicable.

Additional notes

The translation from German was supported by <http://www.DeepL.com/Translator> (free version). Furthermore, we would like to thank the AI tool ChatGPT (OpenAI) for language editing and wording optimization in individual sections under human guidance. All ideas, data and interpretations originate from the authors.

Conflicts of interest

The corresponding author declares on behalf of all authors that there is no conflict of interest.

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